

MAS · Integration Techniques and Applications — Question Patterns and Solution Approaches

Based on: 44 past exam questions from 2016–2025, marking key + exam reports. Original questions in [topics/05_Integration.pdf](#). Numerical examples verified with sympy. Review status: has undergone one round of independent review and was revised according to feedback (see [Review_log.md](#)).

Marking panel reminders

- **The antiderivative of $\ln$ must include absolute value signs** (2022 report). In recent keys this is counted in the "antiderivative" mark, so omitting the absolute value signs may mean the mark is not awarded at all; when the argument is always positive (e.g. $\ln(x^2 + 2)$), the key notes that they may be omitted.
 - **Indefinite integrals must include $+ c$** (almost every partial fractions question has this 1 mark).
 - **Basic algebra skills**: more than 30% of candidates expanded $(\sqrt{3} \tan u + 1)^2$ incorrectly (2018 report); misused the distributive law by writing $\sqrt{2 - \sin^2\theta}$ as $\sqrt{2} - \sin \theta$ (2021 report, F-Q8(c)); wrote the numerator expression incorrectly after finding a common denominator in partial fractions (2024 report).
 - When integrating using a given substitution, **simplify fully** after substituting before integrating (2020, 2023 reports).
 - Volume formulas must include π ; all integral expressions must include dx/dy and the limits (the keys for 2022A-Q17 and 2021A-Q13 combine "limits + correct notation" for 1 mark).
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Pattern 1: Definite integrals with a given substitution ★ CF tested almost every year (from 2016–2025, only 2022 did not test it), 4–7 marks

Fixed five steps (the key's 5 marks are basically these five steps)

1. Find dx from the substitution: $u = 1 - x \Rightarrow dx = -du$; $x = 2 \sin \theta \Rightarrow dx = 2 \cos \theta d\theta$.
2. **Change the limits**: $x = 0 \rightarrow u = 1$, $x = 1 \rightarrow u = 0$.
3. Write the integrand completely in terms of the new variable and simplify (commonly $1 - \sin^2\theta = \cos^2\theta$, $1 + \tan^2u = \sec^2u$).
4. Find the antiderivative.
5. Substitute the limits and give the exact value.

Three types of substitution

- Linear or radical: $u = 1 - x$, $u = \sqrt{x + 2}$, $x = 119u + 1$ (2020F-Q7, 2021F-Q3, 2023F-Q3, 2024F-Q3).
- Power-function substitutions (reverse chain rule): $u = \sin 2x$, $u = \cos 2x$ (2016F-Q5(a), 2018F-Q5).
- **Trigonometric substitution**: $x = a \sin \theta$ for $\sqrt{a^2 - x^2}$; $x = a \tan u$ for $\sqrt{x^2 + a^2}$.

Trigonometric substitution example (2019F-Q6): $\int_0^{\sqrt{3}} \sqrt{1 - x^2/4} dx$, let $x = 2 \sin \theta$; the limits become $0 \rightarrow \pi/3 \Rightarrow \int_0^{\pi/3} 2 \cos^2\theta d\theta = \int (1 + \cos 2\theta) d\theta = \pi/3 + \sqrt{3}/4$.

Past exam questions: 2016F-Q5, 2017F-Q3, 2018F-Q5, 2018F-Q9, 2019F-Q6, 2020F-Q7, 2021F-Q3 (you must choose the substitution yourself), 2021F-Q8(c), 2023F-Q3, 2023F-Q7(b), 2024F-Q3, 2025F-Q6. **Indefinite integral version** (2016A-Q9, $\int x(1 + x)^n dx$): there is no step for changing the limits, and you must substitute back to x at the end.

Pattern 2: Integration after simplifying with trigonometric identities (CF 3–5 marks)

Common identities

- $\cos^2x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2x = \frac{1}{2}(1 - \cos 2x)$;
- $(\sin x + \cos x)^2 = 1 + \sin 2x \Rightarrow \int_0^{\pi/2} \{ \dots \} = \pi/2 + 1$ (2022F-Q3);

- Product-to-sum: $2 \sin A \cos B = \sin(A + B) + \sin(A - B) \Rightarrow \int_0^{\pi/2} 6 \sin(5x/2) \cos(x/2) dx = 4$ (2019F-Q1);
- $\tan^2 x = \sec^2 x - 1$ (2016F-Q5(b));
- $\int_0^{\pi} (4\cos^2 x - \sin x) dx = \int (2 + 2\cos 2x - \sin x) dx = 2\pi - 2$ (2020F-Q1).

Past exam questions: 2016F-Q5(b), 2019F-Q1, 2020F-Q1, 2022F-Q3.

Pattern 3: Partial fractions ★ CF high-frequency (tested in 8 of 10 years, not tested in 2017 or 2023)

Standard procedure

1. Set up the form:
 - Distinct linear factors: $A/(x - a) + B/(x - b)$;
 - **Repeated root**: $A/(x + 1) + B/(x + 1)^2$ (2024F-Q4); $A/x + B/x^2 + C/(x + 3)$ (2019F-Q3);
 - **Irreducible quadratic**: $A/(x - 2) + (Bx + C)/(x^2 + 2)$ (2021F-Q5; $x^2 + 4$ in 2020F-Q6).
 - When the question specifies the form, follow it: 2022F-Q4 requires writing it as $a/(x - 1) + (bx + c)/(x^2 + 3x + 1)$ (this quadratic can actually factorise, since its discriminant is $5 > 0$).
2. After finding a common denominator, equate the numerators, substitute special values or compare coefficients, and find the constants ("equivalence of numerators" 1 mark, solving for the constants 1 mark).
3. Integrate term by term: $A/(x - a) \rightarrow A \ln|x - a|$; $B/(x + 1)^2 \rightarrow -B/(x + 1)$; for terms such as $(2x + 3)/(x^2 + 3x + 1)$ (where the numerator is exactly the derivative of the denominator) $\rightarrow \ln|x^2 + 3x + 1|$; $q/(x^2 + 4) \rightarrow (q/2) \tan^{-1}(x/2)$ (obtained from the derivative of inverse trigonometric functions, 2020F-Q6).
4. Write + c.

Example: $1/(x^2 - 1) = \frac{1}{2}[1/(x - 1) - 1/(x + 1)] \Rightarrow \int = \frac{1}{2} \ln|(x - 1)/(x + 1)| + c$ (2018F-Q6). $(5x + 3)/(x + 1)^2 = 5/(x + 1) - 2/(x + 1)^2 \Rightarrow \int = 5 \ln|x + 1| + 2/(x + 1) + c$ (2024F-Q4). **When no hint is given** (2025F-Q3): first factorise the denominator yourself (the first mark in the key).

Past exam questions: 2016F-Q4, 2018F-Q6, 2019F-Q3, 2020F-Q6(c)(d), 2021F-Q5, 2022F-Q4, 2024F-Q4, 2025F-Q3.

Pattern 4: Area (integrating with respect to x or y)

Approach

1. Find intersection points or points of tangency to use as limits.
2. **Choose the variable of integration**: when the curve is more conveniently written as $x = g(y)$, integrate with respect to y ($2x = \sin y$ in 2016A-Q13; 2017A-Q12; 2022F-Q7). The key often awards marks for both methods, but integrating with respect to y usually requires only one integral.
3. Area = $\int (\text{right} - \text{left}) dy$ or $\int (\text{top} - \text{bottom}) dx$.
4. Use symmetry: ellipse 2025A-Q8; 2022A-Q13(c) multiply by 2 due to symmetry about the x-axis; the 4 in the cardioid 2021F-Q8 = (the 2 from symmetry about the y-axis) $\times$ (the 2 in the difference between the upper and lower branches, $2\sqrt{2 - x^2}$).
5. Interpret a definite integral as an area: 2017A-Q16 (after substitution $u = 2x - a$ it equals the area of some region), 2024A-Q15(c) (triangle area = $2k$).
6. Area of a circular segment: solve the circle equation for $y = 4 \pm \sqrt{4 - (x - 3)^2}$; the difference between the upper and lower branches = $2\sqrt{\dots}$ (2023F-Q7(a)).
7. Determine whether a curve is a circle: compare the numerical area with πr^2 (2020A-Q16(b)).

Past exam questions: 2016A-Q13, 2017A-Q12, 2017A-Q16, 2022A-Q13(c), 2024A-Q15(c), 2018A-Q15(b), 2020A-Q16, 2021F-Q8, 2022F-Q7, 2023A-Q18(b), 2023F-Q7, 2024F-Q8(e), 2025A-Q8.

Pattern 5: Volume of solids of revolution ★ CA high-frequency

Formulas

- About the x-axis: $V = \pi \int_a^b y^2 dx$; about the y-axis: $V = \pi \int_a^b x^2 dy$ (first write x^2 as a function of y).
- Region between two curves: $V = \pi \int (y_{\text{out}}^2 - y_{\text{in}}^2) dx$ (not $\pi \int (y_{\text{out}} - y_{\text{in}})^2 dx$).
- Annulus (doughnut, 2024A-Q11): $x = 3 \pm \sqrt{4 - y^2} \Rightarrow V = \pi \int (R^2 - r^2) dy = \int_{-2}^2 12\pi \sqrt{4 - y^2} dy$.

Contextual question types

- Volume formula for a cup: derive using an integral with letters a , b , h (2017F-Q8).
- Depth when filled to 80%: set $\int_0^h \pi x^2 dy = 0.8 \times V_{\text{full}}$, solve for h (2020A-Q9).
- Related rates for filling with water: $dV/dt = (dV/dh) \cdot (dh/dt)$ (2025A-Q14(b): find dV/dh from $V = ah^3$).
- Increment formula: $\delta h \approx \delta V \div (dV/dh)$, where $dV/dh = \pi x^2|_{y=h}$ (2023A-Q17(c)).
- The volume integral in 2022A-Q17 is in (a); (b)–(d) are about increments, flow rate and separation of variables for $dV/dt = 300e^{-V/12000}$ (see the 'Rates of Change and Differential Equations' handout).
- Spherical cap (2021A-Q13): from the circle equation get $y^2 = 20x - x^2$, integrate with respect to x from 0 to h .

Marking points: ✓ express the radius in terms of x or y ✓ correct integral expression (limits + π + square) ✓ calculation (including units, rounding). **Past exam questions:** 2016A-Q17(c), 2017F-Q8, 2018A-Q15(c), 2019F-Q8, 2020A-Q9, 2021A-Q13, 2022A-Q17, 2023A-Q17, 2024A-Q11, 2025A-Q14.